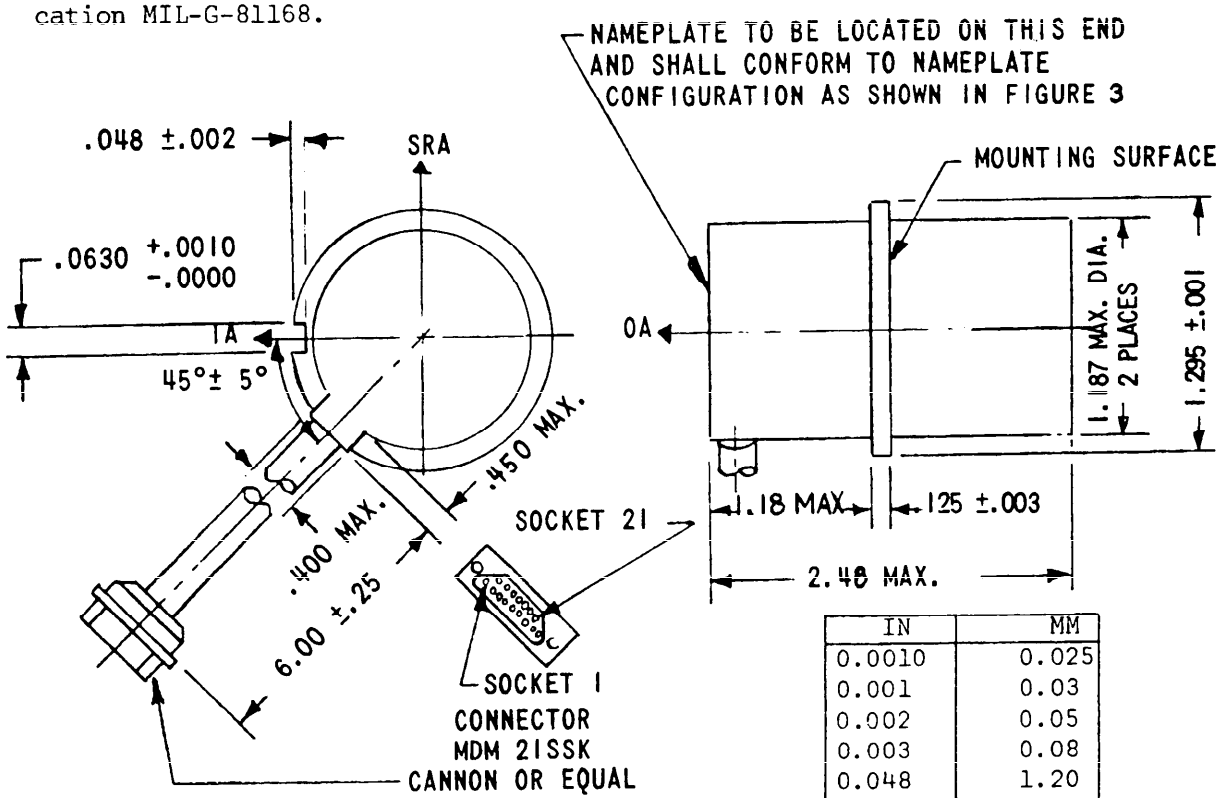
MILITARY SPECIFICATION SHEET

GYROSCOPE, RATE INTEGRATING

The complete requirements for procuring the gyroscope described herein shall consist of this document and the latest issue of specification MIL-G-81168.



IN	MM
0.0010	0.025
0.001	0.03
0.002	0.05
0.003	0.08
0.048	1.20
0.0630	1.600
0.125	3.18
0.25	6.3
0.400	10.15
0.450	11.45
1.18	30.0
1.187	30.15
1.295	32.90
2.48	63.0
6.00	152.4

FIGURE 1. DIMENSIONS AND CONFIGURATION

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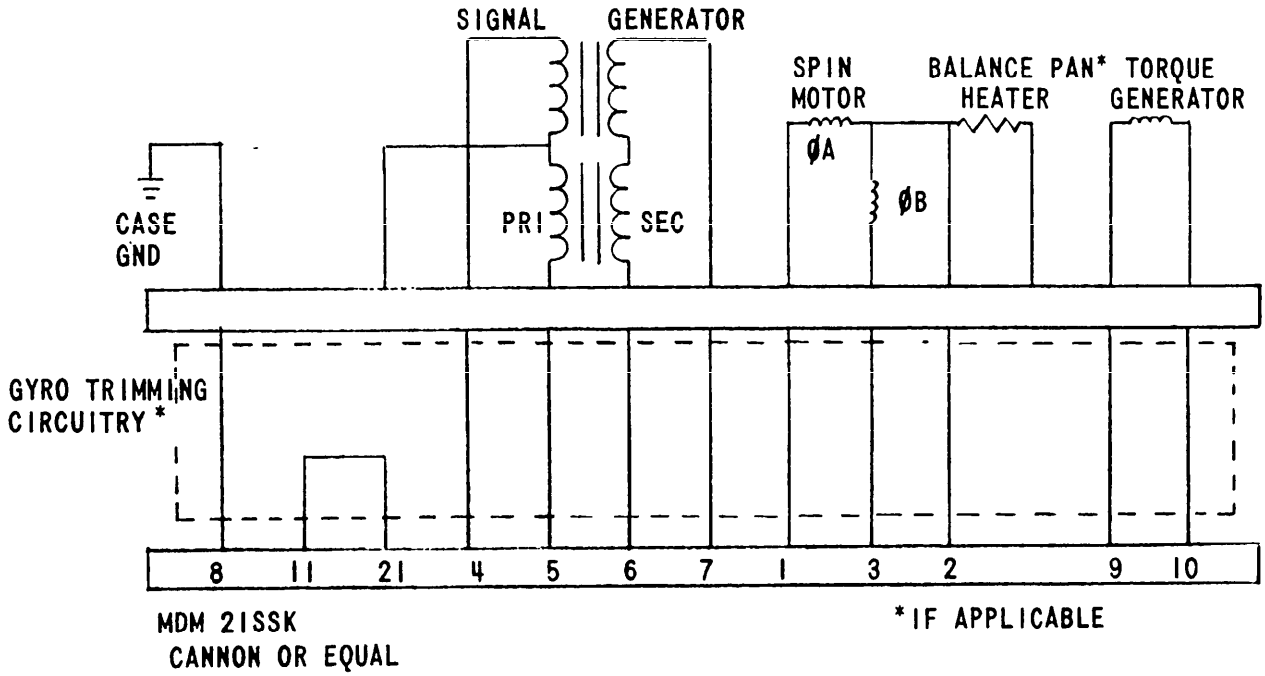


FIGURE 2. SCHEMATIC DIAGRAM

Requirement		Value	Units	Tolerance
Dielectric Strength		250	volts, rms @60 Hz	minimum
Using Test Point Pairs:	1 and case 9 and case 5 and case 6 and case 5 to 6			
Insulation Resistance		100	megohms	minimum
Using a test potential of: Using test point pairs same as for dielectric.		250	volts, D.C.	±10 volts
Spinmotor				
Type:		synchronous, hysteresis		
Winding Configuration:		2 Phase		
Rated Excitation:		15.0	volts, peak-peak sq. wave centered at 0 volts	±1.0V
		850	Hz	Nominal

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Requirement	Value	Units	Toleranc
Spinmotor (cont.)			
Test Excitation*:	15.0	volts, peak-peak sq. wave centered at 0 volts	±.1V
	850	Hz	±.1%
Rise Time (10%-90%)	100	microseconds	maximu
Angular Velocity:	25,500 (425)	rpm (rps)	nomina
Rotor Angular Momentum:	1.32×10^4 (1.32×10^{-3})	gm-cm^2/sec ($kg \cdot m^2/s$)	nomina
Run-up Time to Synchronous Speed:	45	seconds 45	maximu
Run-up Time to 1/2 Synchronous Speed:	15	seconds	maximu
Power Consumption:			
Starting:	5.0	watts	maximu
Running:	4.0	watts	maximu
Power Factor at Synchronous Speed:	0.5		±.1
Starting Current (per phase):	600	ma	maximu
Running Current (per phase):	450	ma	maximu
DC Resistance (per winding @70°F (21°C)):	9.5	ohms	±20%
Impedance (over temperature range -20° to 150°F (-29° to 65.5°C)):	17+j26	ohms	±30%
Pickoff			
Rated Excitation:	20	volts, rms	±10%
	3400	Hz	nomina
Test Excitation*:	20	volts, rms	±1%
	3400	Hz	±.1%
Distortion:	2	%	maximu
Primary Current:	40	ma	nomina
	50	ma	maximu
DC Resistance at 70°F(21°C)			
Primary:	90	ohms	±20%
Secondary:	325	ohms	±20%
Impedance:			
Primary with open secondary:			
@70°F (21°C)	170+j490	ohms	±20%
@-20° to 150°F (-29° to 65.5°C)	170+j490	ohms	±40%
Secondary with open primary:			
@70°F (21°C)	405+j100	ohms	±30%
@-20° to 150°F (-29° to 65.5°C)	405+j100	ohms	±50%

*NOTE: Spinmotor and pickoff excitation shall be supplied from a phase locked frequency source.

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Requirement	Value	Units	Tolerance
Pickoff (cont.)			
Null Voltage:	15	mv, rms	maximum
Phase Shift:	0	degrees	±7 degrees (±120 mrad)
Scale Factor:	31.5	mv/mr	nominal
Linearity Error: When measured at input angles from 0 to ±1.5°(±26mrad) or limit of input angular freedom (whichever is less) in increments of 0.5°(9 mrad).	±5	percent	maximum
Torquer			
Type:	D.C. permanent magnet		
Control Current Range:	0-220	ma	minimum
Control Winding Resistance:			
@70°F(21°C)	160	ohms	±20%
@150°F(65.5°C)	228	ohms	maximum
Control Winding Inductance:	4.0	millihenrys	maximum
Control Winding Time Constant (L/R):	50.0	microseconds	maximum
Command Rate Scale Factor* tolerance from specified value:	Command Rate (rad/sec)	Tolerance (percent)	
	up to 1.0	±.75	maximum
	up to 1.5	±1.5	maximum
	up to 2.2	±5.0	maximum
Command Rate Scale Factor* tolerance from specified value after 4 years:	up to 1.0	+.75 -.95	maximum
	up to 1.5	+1.5 -1.7	maximum
	up to 2.2	+5.0 -5.2	maximum
Maximum Command Rate:	2.25	rad/sec	minimum
Gyro Unit			
Command Rate Scale Factor*:	1.0 (1.745)	deg/sec/ma (mrad/s•ma)	(See Above)
Gyro Scale Factor:	16.6	mv/mr	±25%
Input Angular Freedom:	±0.5 (8.5)	deg (mrad)	minimum
	±2.0 (35)	deg (mrad)	maximum

*NOTE: Command rate scale factor shall be defined as the ratio of output angular velocity to torquer current.

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Requirement	Value	Units	Toleranc
Gyro Unit (cont.)			
Angular Acceleration Sensitivity about the Output Axis:	519 (9.05)	$\frac{deg/hr}{rad/sec^2}$ $\frac{(rad/hr)}{(rad/sec^2)}$	maximu

Gyro Phasing:

Spinmotor:

With a two phase excitation and phase A leading phase B by 90°, positive spinmotor rotation shall occur with terminal 1 connected to phase A and terminal 3 connected to phase B.

Pickoff:

Positive output voltage shall occur when the voltage from terminal 6 to terminal 7 is in phase with the voltage from terminal 5 to terminal 4.

Torquer:

Positive torque shall occur when a positive voltage is applied to termina 10 with respect to terminal 9.

Characteristic Time:			
@-20° to +150°F (-29° to 65.5°C)	2	msec	maximu
@70°F (21°C)	1	msec	nomina
H/J (ratio of wheel momentum to output axis inertia):	530	1/sec	nomina

Axis Alignment:

OA orthogonal to mounting flange surface within 1.5 milliradians. Uncertainty in alignment due to pivot/jewel clearance shall not exceed 0.25 mr.

Input axis aligned to within 1.5 milliradians of the IA notch.

Drift Rate: (-20° to 150°F (-29° to 65.5°C) unless otherwise indicated)

Total = Ri $+\sqrt{(MUS)^2 + (MUI)^2}$:	50 (870)	deg/hr (mrad/hr)	maximu
Acceleration Insensitive (Ri):	21 (370)	deg/hr (mrad/hr)	maximu
Mass Unbalance (MUS, MUI):	30 (525)	deg/hr/g (mrad/hr·g)	maximu
Total Acceleration Sensitive (Resultant):	42 (735)	deg/hr/g (mrad/hr·g)	maximu

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Requirement	Value	Units	Tolerance
Gyro Unit (cont.)			
Drift rate (cont.):			
Random Drift:(30 min. test, max p-p variation commencing 5 minutes after start-up) @135°F (57°C)	6 (105)	deg/hr (mrad/hr)	maximum
Anisoelasticity: (1-500 Hz, 15 g (147 m/s^2) peak)	0.4 (7)	deg/hr/g^2 (mrad/hr·g^2)	maximum
Day-to-day stability:			
g insensitive	6.0 (105)	deg/hr (mrad/hr)	maximum
mass unbalance	6.0 (105)	deg/hr/g (mrad/hr·g)	maximum
Drift Boundaries over Gyro Life:			
Total:	50 (870)	deg/hr (mrad/hr)	maximum
Random:	8 (140)	deg/hr (mrad/hr)	maximum
Elastic Restraint Coefficient:	2.5 (44)	deg/hr/mr (mrad/hr·mrad)	maximum
Input axis rate capability: Maximum sustained rate about IA without damage	20	rad/sec	minimum
Rate Noise:			
Low frequency: Using a first order torque to balance loop with an open loop bandwidth of:	60 (1.05)	deg/hr, rms (rad/hr)	maximum
with no filtering of loop output frequencies below:	200	rad/sec	minimum
	250	rad/sec	minimum
High frequency: Using a first order torque-to balance loop with an open loop bandwidth of:	700 (12.2)	deg/hr, rms (rad/hr)	maximum
with no filtering of loop output frequencies below:	200	rad/sec	minimum
	1000	rad/sec	minimum
Maximum allowed filtering in both cases:	40	dB/decade	maximum

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Requirement	Value	Units	Tolerance
Gyro Unit (cont.)			
Ambient Temperature: (Gyro shall meet all requirements over this temperature range unless otherwise indicated)	-20 to +150 (-20 to 65.5)	°F (°C)	range
Storage Temperature:	-65 to +190 (-54 to 88)	°F (°C)	range
Operating Life:	3000	operating hrs	minimum
or (whichever occurs first):	4	years	minimum
Vibration, shock, altitude and moisture resistance shall be conducted in accordance with MIL-G-81168.			

Markings/Nameplate:

The gyro shall be identified by marking in accordance with MIL-STD-130. All markings shall withstand the same environmental tests required of the gyro and shall remain legible throughout the operating life of the gyro.

The name plate configuration shall be as shown in figure 3. Nameplates shall conform to the requirements of MIL-P-15024, type A, B, C, E or G as appropriate.

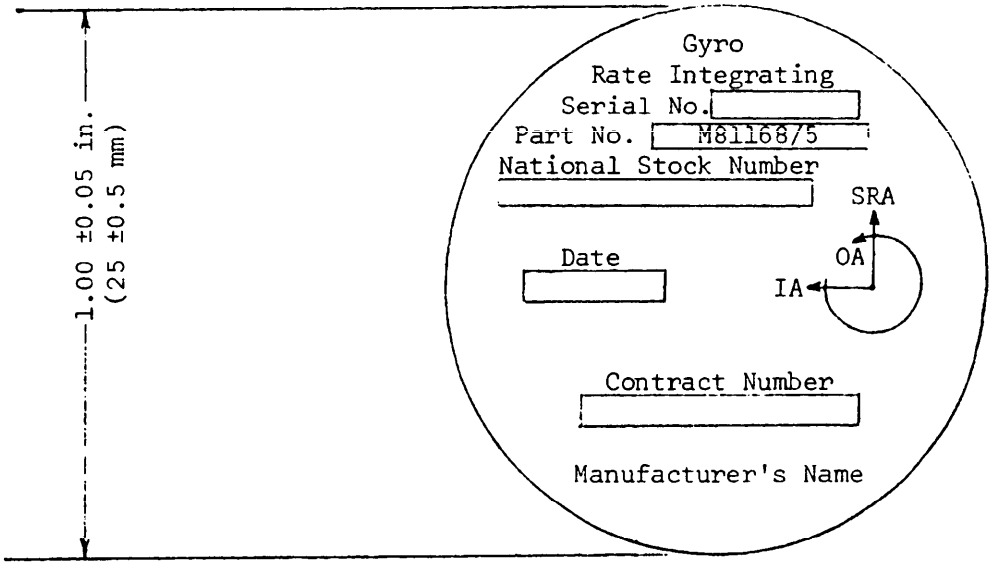


FIGURE 3. NAMEPLATE CONFIGURATION

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Tests and Inspections

The gyro shall meet requirements of this specification when tested and inspected in accordance with the procedures described in MIL-G-81168 and itemized in sequence in Table I therein, with the following exceptions and additions:

a. The following test procedures are not applicable: operating temperature, torquer null voltage, torquer linearity error, temperature sensor inducted voltage, hysteresis, reaction torque, temperature sensor scale factor, and null repeatability. The pickoff impedance shall be measured at 3400 Hz instead of 1000 Hz as indicated in MIL-G-81168.

b. The acceleration insensitive and mass unbalance caused drift rates shall be determined by the six-position servo turntable test procedure. Anisoelasticity shall be determined by the "high-gravity" vibrating table test procedure. These drift rate test procedures are included in MIL-G-81168, section four. Random drift rate shall be determined with the gyro in a servo rate loop, monitoring the torquer current for 30 minutes.

c. The following test procedure, to determine the sensitivity to angular acceleration about the output axis, shall be performed after the characteristic time test procedure. This test shall be performed only during first article inspection.

The gyro shall be mounted on a single-axis oscillating table with a means of driving it sinusoidally at known variable frequencies. The gyro shall be positioned with the output axis parallel to the table axis and shall be energized and stabilized as described in MIL-G-81168. The gyro pickoff shall be monitored by a device capable of reading the peak-to-peak output voltage and the torquer current shall be adjusted to hold the gyro at null. The table shall be operated at a frequency approximately equal to, but not greater than one tenth of the inverse of the gyro characteristic time at an amplitude great enough to produce easily discernible output voltages without causing the gimbal to reach the stops.

The output axis sensitivity is the ratio of the peak-to-peak output voltage of the gyro pickoff to the peak-to-peak amplitude of the table rate divided by the gyro's transfer function.

d. The following test procedure, to determine rate noise, shall be performed after the random drift rate test procedure. This test shall be performed during first article inspection, periodic production inspection, and quality conformance inspection.

The gyro shall be mounted in the test fixture with the output axis vertical and the input axis east and the gyro pickoff shall be connected to the input of a servo rate loop as described in MIL-G-81168. The servo rate

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loop shall have the bandwidth specified herein. The gyro shall be excited and stabilized in accordance with MIL-G-81168. The total rms value of the torquer feedback current above 2 Hz in the frequency range specified in the detail specification shall be determined.

Rate noise is the rms current multiplied by the command rate scale factor of the gyro.

e. In addition to tests at room ambient temperature, the following tests shall be performed during first article inspection and periodic production inspection:

(1) Run-up time at -20°F and 150°F (-29° and 65.5°C)
(2) Command rate scale factor at -20°F and 150°F (-29° and 65.5°C)
(3) Gyro Scale Factor at -20°F and 150°F (-29° and 65.5°C)
(4) Characteristic Time at -20°F and 150°F (-29° and 65.5°C)
(5) Acceleration sensitive and acceleration insensitive drift at -20°F, 135°F, and 150°F (-29°, 57°, and 65.5°C)
(6) Random drift at 135°F (57°C)

f. In addition to impedance tests required by MIL-G-81168, spin motor and pickoff DC resistance and spin motor impedance shall be measured during first article or qualification inspection and periodic production inspection.

NOTE: In accordance with ASTM E 380-72, this specification has been annotated parenthetically with SI (International System of Units) equivalent quantities where applicable non-SI units are used. This was done to illustrate and encourage the use of SI quantities.

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MIL-G-81168/5B(OS), GYROSCOPE, RATE INTEGRATING

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
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